Prog 9 :

Step 1 : create 2 files namespace.yaml and pods.yaml

```
rv24mc028_chethanacs@Elite:~/DEVops/prog9$ cat namespace.yaml
apiVersion: v1
kind: Namespace
rv24mc028_chethanacs@Elite:~/DEVops/prog9$ cat pods.yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
```

Step 2: start minikube and check the status :

```
rv24mc028_chethanacs@Elite:~/DEVops/prog9$ minikube start
 minikube v1.37.0 on Ubuntu 22.04
 Using the docker driver based on existing profile
 Starting "minikube" primary control-plane node in "minikube" cluster
 Pulling base image v0.0.48 ...
 Restarting existing docker container for "minikube" ...
 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
 Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
 Enabled addons: storage-provisioner, default-storageclass
 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
rv24mc028_chethanacs@Elite:~/DEVops/prog9$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 3 : Apply for both namespace.yaml and pods.yaml

```
rv24mc028_chethanacs@Elite:~/DEVops/prog9$ kubectl apply -f namespace.yaml
namespace/dev created
namespace/prod created
rv24mc028_chethanacs@Elite:~/DEVops/prog9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
```

Step 4 : get pods for both

```
rv24mc028_chethanacs@Elite:~/DEVops/prog9$ kubectl get pods -n dev
NAME           READY   STATUS              RESTARTS   AGE
backend-pod    0/1     ContainerCreating   0          9s
frontend-pod   0/1     ContainerCreating   0          9s
rv24mc028_chethanacs@Elite:~/DEVops/prog9$ kubectl get pods -n prod
NAME           READY   STATUS              RESTARTS   AGE
backend-pod    0/1     ContainerCreating   0          18s
frontend-pod   0/1     ContainerCreating   0          18s
```

Step 5 : verify with all the pods

```
rv24mc028_chethanacs@Elite:~/DEVops/prog9$ kubectl get pods
NAME                         READY   STATUS      RESTARTS   AGE
backup-job-29445750-ffmqv    0/1     Completed   0          6d23h
backup-job-29455775-dn5mq    0/1     Completed   0          64s
backup-job-29455780-64gg9    0/1     Completed   0          31s
my-cron-29445750-226zg       0/1     Completed   0          6d23h
my-cron-29455779-xsml6       0/1     Completed   0          64s
my-cron-29455780-zm9hp       0/1     Completed   0          31s
```

Step 6 : filter with labels

```
rv24mc028_chethanacs@Elite:~/DEVops/prog9$ kubectl get pods -n dev -l app=frontend
NAME           READY   STATUS              RESTARTS   AGE
frontend-pod   0/1     ContainerCreating   0          35s
rv24mc028_chethanacs@Elite:~/DEVops/prog9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS              RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0          42s
```